

Yusuf Gadelrab

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Authorized to work in the U.S. for any employer without sponsorship (U.S. Lawful Permanent Resident).

EDUCATION

San Jose State University | San Jose, CA

B.S. Computer Science, Applied Mathematics track | Expected May 2028

- **Coursework:** Data Structures & Algorithms, Object-Oriented Design, Computer Architecture, Discrete Mathematics, Linear Algebra, Probability & Statistics.
- **Lynbrook High School — Valedictorian**, 4.0 GPA (2024).

EXPERIENCE

HwyHaul | San Jose, CA

AI / Agent Engineering Intern | May - July 2026

- Built a WhatsApp load-intake agent in **FastAPI** that parses unstructured carrier messages into structured TMS records, **cutting entry time from ~90s to under 2s per load**.
- Engineered a local-first extraction pipeline (**Ollama / Qwen-3** with a deterministic regex fallback) so the product **degraded gracefully instead of failing** when the model was unavailable.
- Shipped the pilot behind a full **pytest** suite and **Alembic** schema migrations; demoed the end-to-end carrier onboarding and billing flow to the CEO and operations lead.

IBM SkillsBuild — AI Experiential Learning Lab | Remote

NLP & Data Science Project Lead | Jan - May 2026

- Built an end-to-end NLP pipeline on **IBM Watson** that scores equities from financial-news sentiment, reaching **78% directional accuracy** on a six-month backtest.
- Automated ingestion of **50+ live market sources daily** into a scoring pipeline, **cutting manual research time 60%**.
- Applied supervised and unsupervised models to structured financial data; earned IBM AI Fundamentals credentials in NLP, model evaluation, and responsible AI design.

SJSU CSEd Research Lab (Dr. Ethel Tshukudu) | San Jose, CA

Undergraduate Researcher | Aug 2024 - Present

- Co-authored a **peer-reviewed paper accepted to ACM SIGCSE TS 2026** on bilingual CS learning (doi:10.1145/3770761.3777339).
- Designed and analyzed pre/post instruments for an **IRB-approved mixed-methods study of 60 participants**, establishing **statistically significant pre-to-post gains** in programming confidence, computing identity, enjoyment, and motivation, with novices gaining significantly more in confidence than experienced programmers.

Silicon Valley Entrepreneurship Club | San Jose, CA

Technical Ops & Web Lead | Dec 2025 - Present

- Build and maintain the club's full-stack site (sventclub.org) and internal automation tooling, raising monthly visitor engagement **40%**.
- Secured **4 corporate sponsors** and expanded the operating budget **35%** by pitching Silicon Valley firms.

SJSU Coding Warriors & CS Department | San Jose, CA

CS Tutor & Curriculum Lead | Aug 2024 - Present

- Authored the challenge-based Python and Java curriculum now used for weekly sessions, and taught **10+ workshops to 30+ students**.

PROJECTS

DIRA — static application-security scanner | Python (zero-dependency), OSV.dev, CycloneDX/SPDX, SARIF — 138 passing tests

- Shipped a static security-scanner CLI (secrets, dependency CVEs, config/lac/LLM rules, license risk, git-history leaks, live TLS) with SBOM export and an 18-check security-readiness score.
- **Measured precision before shipping rather than claiming it:** audited all **169 findings** across four real repositories — including one rule wrong on **17 of 17** hits, which I narrowed — and fixed two silent false-negative classes that were discarding any credential containing "secret" or "password".
- Re-ran to **84 findings, every one traced to a verified underlying fact**, and hardened the scanner against a hostile fixture tree (minified bundles, symlink loops, invalid UTF-8, 40-level nesting) it now traverses without crashing or hanging.

Prometheus Tutor | Python, knowledge-graph retrieval, LLM explanation engine — 90 passing tests

- Built an adaptive CS tutor that models prerequisite concepts as a graph and routes each learner to the specific **upstream** concept their wrong answer implicates, rather than re-explaining the missed question.

FreightDesk suite | Python (zero-dependency, stdlib HTTP services) — 66 passing tests

- Designed three composable back-office services (carrier intake, dispatch triage, AR billing) around document extraction, each independently testable and demo-servable.

PUBLICATIONS

- Tshukudu, E., Shah, N., Kieu, T., Deeb, L., Venkateswaran, H., Ghai, A., **Gadelrab, Y.**, & Hada, P. (2026). "Exploring Bilingual Coding for Inclusive Computer Science Learning." *SIGCSE TS 2026*, ACM, 2 pages. doi:10.1145/3770761.3777339

SKILLS

- **Languages:** Python, TypeScript/JavaScript, Java, SQL, C; Arabic and English (native).
- **Frameworks & Tools:** React, Next.js, FastAPI, Node.js, Bun, Prisma, pytest, Git, Docker, Ollama, IBM Watson Studio, Unix/Linux.
- **Focus areas:** LLM application engineering, RAG and graph retrieval, NLP, static analysis, test-driven development.